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K-Medoids-based ICA and Wavelet-Enhanced Multi-resolution Convolutional Network for Motor Imagery EEG

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Abstract: In BCI systems, addressing the limitations where raw MI-EEG signals exhibit insufficient representation of task-related features in spatial-temporal and time-frequency domains with heavy reliance on manual intervention, and existing convolutional neural networks struggle to simultaneously capture local EEG features and global long-term neural oscillation patterns, we propose an end-to-end compact deep convolutional neural network. Our model integrates multi-branch convolutional networks with temporal convolutional networks to extract multi-resolution features. By combining KMedoids ICA and discrete wavelet transform, the original EEG signals are converted into a feature matrix composed of multi-layer wavelet coefficients. Experimental results under subject-independent validation demonstrate that our method achieves classification accuracies of 70.08% and 70.16% on BCI Competition IV-2a and Physionet four-class motor imagery datasets respectively, outperforming other classical deep learning models. Feature visualization reveals clearer inter-class separation and higher intra-class consistency compared with alternative approaches.

Key Words: Motor Imagery, Subject-independent, Electroencephalography, Convolutional Network

1 Introduction

A Brain-Computer Interface (BCI) system is a technology that decodes human intent from neural activity and converts EEG signals into control commands of devices. Motor imagery serves as the primary paradigm for EEG-based research [1]. The classification results of motor imagery can be encoded into predefined machine control commands to control external end devices, which aim to accurately recognise the user's motor intentions such as hand, elbow, foot, fist and tongue movements. This has important implications for fields such as medical rehabilitation [2] and prosthetic control [3], but the field still faces a significant problem: the variation of EEG data across individuals. The variation in EEG signals can be attributed to several factors, including distinct brain structures, varying mental states, and individual head shapes [4]. To mitigate these challenges, research in this field is typically categorized into two primary approaches. The first approach involves subject-dependent models that necessitate personalized calibration. Although these models tend to offer higher accuracy, they require ongoing calibration for each new user, resulting in resource-intensive implementations [5][6]. The second approach focuses on subject-independent models, which aim to identify features that are consistent across different individuals, thereby eliminating the need for individualized calibration [7][8]. While this approach requires fewer resources, it frequently results in diminished performance [9]. Therefore, exploring the subject-independent methods for decoding motor imagery is essential and practically necessary to fully leverage the advantages of BCI in various real-world applications.

In recent years, deep learning has demonstrated superior performance in processing complex and non-linear data, driving its widespread use in subject-independent motor imagery electroencephalography signal (MI-EEG) decoding tasks. Among them, Convolutional Neural Networks (CNN) have gained increasing attention in the field of MI-EEG decoding. For example, the study in [10] proposed a deep CNN architecture and a shallow CNN architecture for the MI-EEG decoding task. The authors of [11] integrated separable convolutional operations into CNN and proposed the EEGNet framework, which was successfully applied to the MI-EEG decoding task. The study in [12] proposed the FBCNet architecture, which divides the EEG signal into multiple bands using filter bank CSP and uses CNN to extract features and decode MI-EEG. The authors of [13] proposed the IFNet architecture, which improves the representation of MI-EEG features by further exploring cross-frequency interactions. The authors of developed EEGNeX, which uses a depth-separated convolutional layers using padding and increasing the receptive field of the CNN by reducing the activation layer. Temporal Convolutional Network (TCN) further extends the application of CNN in time series modelling [15]. In contrast to traditional CNN, TCN is capable of significantly expanding the receptive field size while maintaining a linear increase in the number of parameters. Furthermore, unlike RNN, TCN is not prone to the issues of vanishing or exploding gradients. TCN has demonstrated superior performance over other recurrent architectures, such as LSTM and GRU, in various tasks involving sequential data [15]. Recently, TCN-based models have been applied to the decoding of MI-EEG. The authors of [16] designed an EEG-TCNet architecture that integrates TCN with EEGNet. The authors of [17] improved on this with the TCNet-Fusion model by using the TCN structure and the feature fusion techniques to enhance the EEG-TCNet model. The authors of [18] and [21] achieved excellent performance by using attention-integrated TCN and CNN to decode MI-EEG.

However, limitations have been observed in existing deep learning approaches when processing motor imagery EEG signals. First, current deep learning models lack the

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capability for multi-scale modeling of EEG signals. When raw motor imagery EEG signals are directly fed into deep learning models, the insufficient feature representation capability further deteriorates cross-subject classification performance.

Responding to the above limitations, a K-Medoids-based and Wavelet-Enhanced Multi-resolution Convolutional Network (KWEM-Net) is proposed in this paper. The model was developed by introducing K-Medoids-based ICA (KMICA) and Discrete Wavelet Transform (DWT) as preprocessing steps, KMICA for independent component separation and noise suppression of EEG signals, and DWT from time-frequency domain to enhance the input features. The deep network part of KWEM-Net further extracts multi-resolution features using convolution kernels of different sizes and models the sequence properties of the input features by introducing TCN. The rest of the paper is organised as follows. In Section 2, we describe the model proposed in this paper. Section 3 describes the public dataset used for experiments and its experimental results, and Section 4 provides a brief summary.

2 Method

The structure of KWEM-Net is shown in Fig. 1 and consists of three components: KMICA, DWT module and a deep network part. The specific structure of each module and its role will be described in detail in the next section.

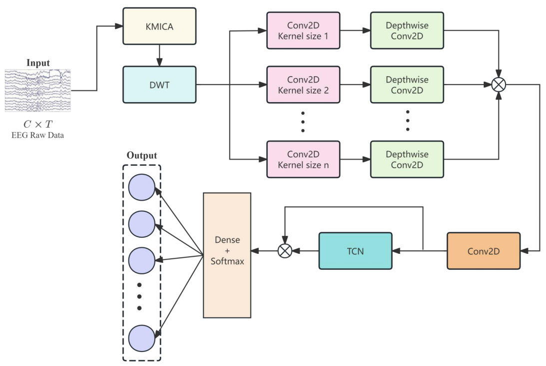

Fig. 1: KWEM-Net structure

2.1 KMICA

Raw MI-EEG signals are often contaminated by spontaneous brain activity, artifacts, and electromagnetic noise, resulting in an extremely low SNR. This significantly hinders deep learning models from effectively capturing discriminative features and degrades classification performance. To enhance signal quality, ICA has been widely adopted in EEG preprocessing, with FastICA being one of the most prevalent algorithms. However, FastICA introduces uncertainty in the order and polarity of independent components due to random initialization, which compromises the quality of signal reconstruction.

To address this limitation, we propose a K-Medoids-Based ICA (KMICA) method by integrating the Monte Carlo principle. KMICA iteratively applies FastICA to extract independent components, employs Pearson correlation coefficients to evaluate component similarity for order alignment, and leverages K-Medoids clustering to determine the centroid of each component category. This framework enables the extraction of stable signal sources and the reconstruction of robust MI-EEG signals.

The workflow of KMICA is shown in Fig. 2. Firstly, the number of independent components extracted is set to be the number of channels of EEG data n, and a source matrix $S^{(i)} \in R^{nxm}$ and a mixing matrix $A^{(i)} \in R^{nxn}$ are obtained from each decomposition. Multiple FastICA decompositions are performed on the input EEG data $X \in R^{nxm}$, and the number of FastICA runs is r. n denotes the number of channels (the number of independent components), and m denotes the number of time samples, and the process can be expressed as:

$$X = A^{(i)}S^{(i)}(i = 1, 2, ..., r-1, r) \tag{1}$$

$$S^{(i)} = \left[s_1^{(i)}, s_2^{(i)}, ..., s_k^{(i)}, ..., s_n^{(i)}\right]^T \tag{2}$$

where $s_k^{(i)}$ denotes the kth row in the source matrix $S^{(i)}$ obtained in the ith run, the kth independent component vector obtained in the ith run. To ensure consistent ordering of independent components across multiple FastICA runs, mu-

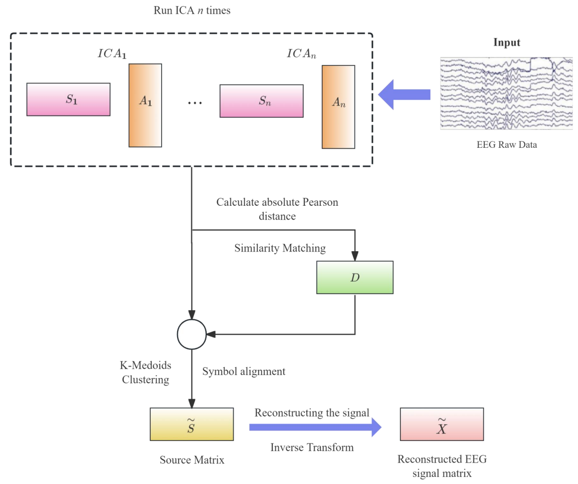

Fig. 2: KMICA workflow

tual information could theoretically be used to evaluate inter-batch correspondences [22]. However, given that only component order alignment is required here, we adopt a more computationally efficient approach. Correlation, serving as a simplified proxy for mutual information, can effectively quantify linear relationships between non-Gaussian signals [23]. We therefore leverage the absolute Pearson distance between independent components to measure their similarity. A similarity-based matching approach is implemented to correct component order variations, thereby achieving standardized alignment of independent component sequences. Specifically, for the kth component vector $s_k^{(i)}$ in the ith run and the lth component vector $s_l^{(j)}$ in the jth run, the distance $d_{absp}(s_k^{(i)}, s_l^{(j)})$ is defined as expressed in the following equations:

$$d_{absp}(s_k^{(i)}, s_l^{(j)}) = 1 - \left|\rho(s_k^{(i)}, s_l^{(j)})\right| \tag{3}$$

$$\rho(s_k^{(i)}, s_l^{(j)}) = \frac{\sum_{u=1}^{m}[s_k^{(i)}(u) - \overline{s_k^{(i)}}][s_l^{(j)}(u) - \overline{s_l^{(j)}}]}{\sqrt{\sum_{u=1}^{m}[s_k^{(i)}(u) - \overline{s_k^{(i)}}]^2} \cdot \sqrt{\sum_{u=1}^{m}[s_l^{(j)}(u) - \overline{s_l^{(j)}}]^2}} \tag{4}$$

where $d_{absp}(s_k^{(i)}, s_l^{(j)})$ is used to measure the similarity between any two independent components $s_k^{(i)}$ and $s_l^{(j)}$, with a value domain of $[0,1]$. the absolute value of the Pearson correlation coefficient is used to measure the linear correlation between $s_k^{(i)}$ and $s_l^{(j)}$.

The above calculations allow for the construction of a symmetric distance matrix *D*. Since the number of FastICA runs is *r*, *n* denotes the number of independent scores, and the dimension of the distance matrix is $rn \times rn$, the specific structure of this matrix is shown in the following equation:

$$D = \begin{bmatrix} D^{(1,1)} & D^{(1,2)} & \cdots & D^{(1,r)} \\ D^{(2,1)} & D^{(2,1)} & \cdots & D^{(2,r)} \\ \vdots & \vdots & \ddots & \vdots \\ D^{(r,1)} & D^{(r,2)} & \cdots & D^{(r,r)} \end{bmatrix} \tag{5}$$

where each chunk matrix $D^{(i,j)}$ represents the distance between all pairs of components in the *i*th and *j*th runs. Specifically, the element $D_{k,l}^{(i,j)}$ of the *k*th row and *l*th column of the chunking matrix $D^{(i,j)}$ is represented as the absolute Pearson distance between $s_k^{(i)}$ and $s_l^{(j)}$:

$$D_{k,l}^{(i,j)} = d_{absp}(s_k^{(i)}, s_l^{(j)}) = 1 - \left|\rho(s_k^{(i)}, s_l^{(j)})\right| \tag{6}$$

where the elements on the diagonal of each chunking matrix $D^{(i,j)}$, the diagonal elements of the distance matrix *D,* are usually 0 because they are the auto-correlation distances of the same component vectors; the off-diagonal elements denote distances between the same or different components in different runs.

Following similarity matching, independent components sharing the same identifier must be clustered. To ensure directional consistency during clustering, sign alignment is first implemented. This critical preprocessing step prevents components with opposing polarities from mutual cancellation during cluster centroid computation, thereby enhancing the stability of the final reconstructed results. Taking the *k*th extracted independent component component as an example, in order to ensure that the sign direction of the obtained independent components in multiple runs is the same, the *k*th independent component $s_k^{(1)}$ of the source matrix $S^{(1)}$ obtained in the first run is used as a reference to align the signs of the independent components $s_k^{(i)}$ of the same order obtained in the subsequent runs one by one in the process shown in the following equation:

$$sign(s_k^{(i)}) = \begin{cases} +1, & corr(s_k^{(1)}, s_k^{(i)}) > 1 \\ -1, & corr(s_k^{(1)}, s_k^{(i)}) < 1 \end{cases} \tag{7}$$

Where $corr(s_k^{(1)}, s_k^{(i)})$ denotes the correlation coefficient between the independent components $s_k^{(1)}$ and $s_k^{(i)}$. In this way, it is ensured that the components in all FastICA runs are consistent in sign, which facilitates the calculation of the subsequent parts.

Based on the distance matrix *D*, the K-Medoids clustering algorithm is used to cluster all independent components into *n* component clusters. In K-Medoids clustering, any cluster of *n* component clusters is denoted by C_k and the selection process of the central component m_k is represented as:

$$m_k = \arg\min_{x \in C_k} \sum_{y \in C_k} D(x, y) \tag{8}$$

Where C_k denotes the *k*th cluster containing all the components that are grouped in the clustering process. m_k is the central component of the cluster that has the smallest sum of distances from all other components in C_k. $D(x,y)$ is the absolute Pearson distance between component *x* and component *y*. For each C_k, the mean of all components in the cluster is calculated to obtain the robust independent component $\widetilde{s}_k$. Finally, the new source is formed as described in the following equation:

$$\widetilde{S} = [\widetilde{s}_1, \widetilde{s}_2, ..., \widetilde{s}_k, .., \widetilde{s}_n] \tag{9}$$

where $\widetilde{S}$ denotes the source matrices, respectively.

The robust independent component matrixs generated through clustering and sign alignment of components obtained from multiple FastICA runs. Aligned with the order and polarity of the first-run results, it maintains close association with the initially derived mixing matrix. This inherent correspondence enables EEG signal reconstruction via inverse transformation of these two matrices.

2.2 DWT

MI-EEG signals organized by electrode channels and temporal dimensions exhibit inherently coupled time-frequency dynamics across neural activation patterns, causing spectral-temporal aliasing that hinders discriminative feature extraction in CNNs [24]. We address this via Discrete Wavelet Transform (DWT), which decouples time-frequency characteristics while preserving critical signal features [25]. DWT enhances discriminability by resolving fine-grained frequency variations and preventing spatiotemporal overlap, outperforming other wavelet methods in computational efficiency without sacrificing resolution [26]. The signal is decomposed into high-frequency detail coefficients and low-frequency approximation coefficients, generating a multi-level wavelet coefficient matrix as network input. The wavelet decomposition can be expressed as:

$$\widetilde{s}_k(t) = \sum_{j=1}^{J} \sum_{l} c_{j,l} \varphi_{j,l}(t) + \sum_{l} c_{A,l} \phi_{J,l}(t) \tag{10}$$

Where $\widetilde{s}_k(t)$ denotes the *k*th component, *J* represents the number of layers of the decomposition, $\varphi_{j,l}(t)$ denotes the wavelet basis function of the *j*th layer to extract the high frequency details of the signal, and $\phi_{J,l}(t)$ denotes the scale basis function on the *J*th scale to extract the low frequency approximation. $c_{j,l}$ and $c_{A,l}$ denote the high frequency detail coefficients and low frequency approximation coefficients, respectively.

2.3 Deep Network

The deep network part takes as input the multi-resolution decomposed feature coefficient matrix output from the discrete wavelet transform module. In the initial part of the network structure, the input features obtained after KMICA and DWT processing are processed by multiple parallel convolution operations, with each parallel branch employing a different size of convolution kernel for channel-by-channel extraction of finer-grained features. In this paper, the number of branches set is 5, and their convolution kernel sizes are 2x1, 4x1, 8x1, 16x1, and 32x1. The subsequent deep convolution serves in combining the feature information from different channels to achieve cross-channel information fusion and thus extracting the interrelationships between brain regions. This is followed by further dimensionality reduction through pooling operations and reducing the risk of over-fitting through regularisation. The features generated from different branches are spliced in feature dimensions to form a global feature representation after completing the processing. The spliced features are further integrated by a convolutional kernel of size 16x1, which is subsequently spread as a one-dimensional vector to model the sequence properties of the input features through the use of a time-ordered convolutional network.The TCN constructs sequence-based contextual relationships by ensuring that each position in the sequence of features is dependent only on the current and previous positional features through multiple layers of causal convolution. After each layer of convolution operation, the convolution output is summed with the initial input features through residual concatenation to ensure the integrity of the information flow and the stability of the network training. The entire deep network part generates high-quality feature representations containing multi-band and sequence properties through a combination of multi-scale feature extraction by multi-branch convolution and sequence pattern modelling by TCN, providing comprehensive feature support for the classification task.

3 Experiments and Results

The public datasets involved in the experiment of this paper include BCI Competition IV-2a [19] and Physionet [20]. The models involved in the evaluation include the following:

(1) DeepConvNet [10]

(2) ShallowConvNet [10]

(3) EEGNet [11]

(4) EEGNeX [14]

(5) EEGTCNet [16]

(6) ATCNet [18]

(7) KWEM-Net (Proposed method)

In this study, method performance is evaluated based on accuracy (ACC), defined as the ratio of correctly classified trials to the total number of trials. The experimental setup is as follows: for hardware, the system runs on Windows 11, with an Intel Core i5-12490F processor and an NVIDIA GeForce RTX 4070 graphics card, ensuring efficient and stable computational performance. For software, Python 3.8 was used as the programming language, with PyCharm as the integrated development environment and TensorFlow 2.8.0 as the deep learning framework. The subject-independent classification experiment in the BCI Competition IV-2a employed the "LOSO" [18]. All models were trained on BCI Competition IV-2a with the following settings: learning rate of 0.0009 and parameter optimisation using the adam algorithm, batch size of 64, training period of 800, early stopping tolerance of 300, and cross-entropy as the loss function.The experiment used in Physionet was a 10-fold cross-validation approach. All models were trained on Physionet using the following settings: the learning rate was 0.0009 and the parameters were optimised using the adam algorithm, the batchsize size was 32, the training period was 800, the tolerance for the early stopping method was set to 300, and the cross-entropy was used as the loss function.

Table 1 shows the subject-independent classification results of KWEM-Net and all baseline methods on BCI Competition IV-2a. It can be seen that KWEM-Net achieves an average accuracy of 70.08% in the LOSO-based subject-independent experiments, taking the lead compared to the other comparison models. The ACC of KWEM-Net is the highest value on the data of subjects A01, A03, A04, A07, A08, and A09. In contrast, DeepConvNet, despite achieving close performance on some subjects (A01, A02, A03, A05, and A08), have a lower overall performance. ATCNet and EEGTCNet, although achieving better results on some subjects (A01, A03, and A08), have an overall performance that cannot be compared to that of KWEM-Net. Table 2 shows the results of KWEM-Net and all baseline methods on Physionet. It can be seen that KWEM-Net significantly outperforms the other compared models with a classification accuracy of 70.16%. ATCNet and EEGTCNet have the next best performance with 67.35% and 67.57%, respectively, suggesting that they have some advantages in feature extraction and time series modelling. EEGNeX, with an accuracy of 67.03%, performs slightly lower than the

Table 1: Results of Accuracy Performance Comparison of Methods in BCI Competition IV-2a

subject	Methods and their Acc (%)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A01	74.83	74.13	70.83	53.99	72.92	74.31	**75.35**
A02	**55.9**	54.34	48.61	55.38	46.35	51.56	55.38
A03	82.99	81.08	79.17	67.36	81.6	82.12	**83.68**
A04	55.21	59.9	52.43	48.61	59.72	59.9	**65.45**
A05	63.37	60.24	52.6	59.72	49.13	**64.58**	62.15
A06	**60.76**	55.56	55.73	58.51	50.17	51.91	55.38
A07	77.26	75.35	75.87	71.7	73.78	75	**80.03**
A08	76.39	79.51	74.31	59.2	77.26	78.12	**78.3**
A09	68.58	72.92	59.72	58.51	70.66	72.05	**75**
mean	68.36	68.11	63.25	59.22	64.62	67.73	**70.08**

Table 2: Results of Accuracy Performance Comparison of Methods in Physionet

Methods	Acc (%)
DeepConvNet	64.54
ShallowConvNet	65.41
EEGNet	66.38
EEGNeX	67.03
EEGTCNet	67.57
ATCNet	67.35
Proposed method	**70.16**

previous two, but still better than EEGNet (66.38%). The ACC of DeepConvNet and ShallowConvNet are 64.54% and 65.41%, respectively.

The above comparative experimental results demonstrate the classification accuracy advantage of two datasets, next the differences in feature extraction capability between the methods are further analysed by t-SNE visualisation. Fig. 3 demonstrates the t-SNE visualization results of different models on the BCI Competition IV-2a dataset when subject 9 serves as the test set. It shows that proposed method exhibits a stronger inter-class separation ability compared to ATCNet, DeepConvNet, EEGNet, EEGNeX, EEGTCNet and ShallowConvNet models, and the ATCNet model also shows some inter-class separation ability, but its There is some overlap of feature points between the feet and tongue tasks, and the overlap is mixed with a large number of right

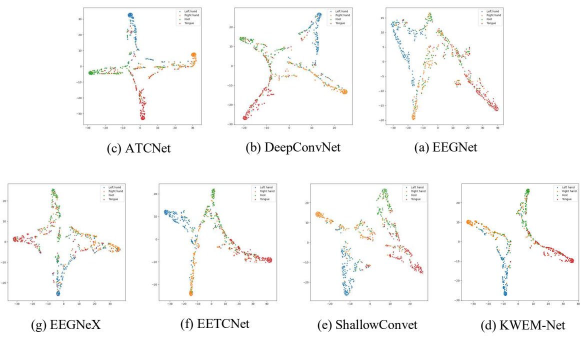

Fig. 3: Feature visualization results of each model when subject 9 of BCI Competition IV-2a is used as the test set

hand samples, and the right hand samples are not aggregated to a high degree. DeepConvNet and EEGTCNet models also perform well in feature extraction, although there is a small amount of confusion between the left-handed and right-handed categories. The results of EEGNet, EEGNeX, and ShallowConvNet show more dispersed feature points, with less clear inter-category boundaries, where there is significant confusion between the left-handed and tongue tasks. Fig. 4 is the t-SNE results of models on Physionet, which shows

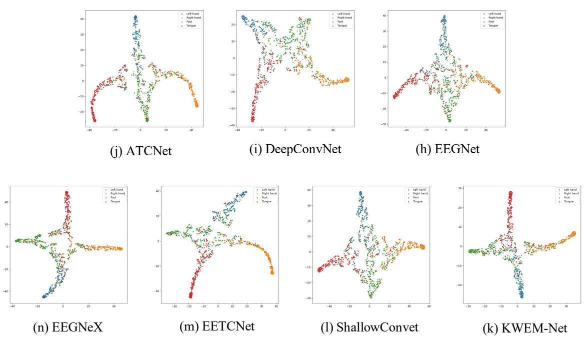

Fig. 4: Results of t-SNE visualisation for each model on Physionet

that KWEM-Net has a clear advantage with higher intra-class consistency and a clearer inter-class separation of four kinds of feature points. In contrast, although ATCNet and EEGTCNet also have better inter-class separation, there is still some overlap in some categories, especially the left-handed and right-handed feature points, while the feature distributions of DeepConvNet and ShallowConvNet are much more dispersed, with more mixing of various types of feature points, and unclear class boundaries, while the feature distributions of EEGNet and EEGNeX feature distributions, although relatively better, especially EEGNeX has improved in inter-class separation, but still cannot reach the compactness and clarity of KWEM-Net. Compared with other models, KWEM-Net forms independent and clear clusters of the four classes of feature points in the low-dimensional space, with higher intra-class consistency and more significant inter-class separation.

4 Conclusion

In this paper, we propose KWEM-Net, a novel end-to-end model integrating K-Medoids-based independent component analysis and discrete wavelet transform. We evaluate its efficacy through cross-subject classification experiments on two benchmark datasets: BCI Competition IV-2a and the PhysioNet EEG Motor Movement/Imagery Dataset. Experimental results demonstrate that KWEM-Net achieves a mean accuracy of 70.08% on BCI Competition IV-2a, outperforming baseline methods. Superior performance is observed for subjects A01, A03, A04, A07, A08, and A09. On the PhysioNet dataset, the model attains a competitive accuracy of 70.16%. t-SNE visualizations further validate enhanced inter-class separability and intra-class consistency in feature distributions. Therefore, the above demonstrates the effectiveness of the model proposed in this paper.

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